

Onyx — FileKit reference

Onyx · multi-process OS on Raspberry Pi 4

The reference of **FileKit** (`SD:/Lib/filekit.so`, `user/Kits/filekit`): what it is for, how a program uses it, and every operation it exposes. The operations' part is made from the kit's headers by `tools/docgen/kitdocs.py` — the headers are the source. Overview of all the kits: [The Kits](#).

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What it is

FileKit is files and folders: whole files read and written, trees copied, moved and removed, paths, compression (zlib), and archives — ZIP read and written, tar / tar.gz / gzip read.

Include	<code>#include "filekit/filekit.h"</code>
Link	<code>lib/filekit.imp.a</code>
Library	<code>SD:/lib/filekit.so</code> — 78 entries in its table (<code>user/Kits/filekit/filekit.abi</code> , append-only)
Sources	<code>user/Kits/filekit/</code>

Using it

A whole file, read and written:

```
#include "filekit/filekit.h"

void *data; unsigned n;
if (fk_load ("SD:/docs/notes.txt", &data, &n) == 0)
{
    /* data: n bytes, a 0 after them */
    fk_save ("SD:/docs/notes.bak", data, n);
    fk_free (data);
}
```

Folders and paths:

```
#include "filekit/filekit.h"

char path[FS_PATHL];
fs_join (path, sizeof path, "SD:/docs", "report.txt");
```

```
if (!fs_exists (path)) { /* ... */ }
```

```
fk_mkdirs ("SD:/backup/2026/docs");           // every folder of the path
fk_copy ("SD:/docs", "SD:/backup/2026/docs", 0, 0); // a file, or a folder and all it holds
fk_remove ("SD:/tmp/work");                     // the same, removed
```

A ZIP archive, read:

```
char err[80];
fk_zip *z = fk_zip_open ("SD:/downloads/pack.zip", err, sizeof err);
if (z)
{
    struct fk_zip_entry e;
    for (int i = 0; i < fk_zip_count (z); i++)
        if (fk_zip_entry (z, i, &e)) ax_putln (e.name);
    fk_zip_extract_all (z, 0, "SD:/downloads/pack", 0, 0);
    fk_zip_close (z);
}
```

A ZIP archive, written:

```
fk_zipw *w = fk_zipw_create ("SD:/backup/docs.zip");
fk_zipw_add (w, "SD:/docs", 0); // the folder and all it holds
fk_zipw_close (w, 0, 0, err, sizeof err);
```

Other archive formats (tar, tar.gz, gzip) are read through the same calls (`fk_arc_*`); a program asks the library which formats it handles (`fk_arc_formats`) instead of keeping its own list.

Index

Everything the headers declare, in their order — the details are in each header's part below.

Name	What it does	Header
<code>fk_free</code>	a buffer the library returned	<code>filekit.h</code>
<code>fk_crc32</code>	(crc: 0 at first, then the last result)	<code>filekit.h</code>
<code>fk_adler32</code>	(adler: 1 at first)	<code>filekit.h</code>
<code>fk_deflate</code>	The whole of data compressed (level 0..9, -1)	<code>filekit.h</code>
<code>fk_inflate</code>	...	<code>filekit.h</code>
<code>fk_zip_entry</code>	(a type)	<code>filekit.h</code>
<code>int</code>	A step of a long work	<code>filekit.h</code>
<code>fk_zip_open</code>	0: err says why	<code>filekit.h</code>
<code>fk_zip_close</code>		<code>filekit.h</code>
<code>fk_zip_error</code>	the last failure's words	<code>filekit.h</code>
<code>fk_zip_password</code>	for its encrypted entries	<code>filekit.h</code>
<code>fk_zip_count</code>		<code>filekit.h</code>
<code>fk_zip_find</code>	its index (letters' case ignored), -1	<code>filekit.h</code>

Name	What it does	Header
<code>fk_zip_read</code>	entry i in memory (fk_free) -> 0 / -1	<code>filekit.h</code>
<code>fk_zip_extract</code>	entry i as that file -> 0 / -1	<code>filekit.h</code>
<code>fk_zip_extract_all</code>	Everything under prefix ("" or 0	<code>filekit.h</code>
<code>fk_zipw_create</code>		<code>filekit.h</code>
<code>fk_zipw_add</code>	a file, or a folder and all it holds, as name (0: its own)	<code>filekit.h</code>
<code>fk_zipw_add_data</code>	a buffer as an entry (copied)	<code>filekit.h</code>
<code>fk_zipw_level</code>	0 store, 1 fast, 6 (the default), 9	<code>filekit.h</code>
<code>fk_zipw_close</code>	written -> 0 / -1 (err); w freed	<code>filekit.h</code>
<code>fk_format</code>	The formats are asked, not assumed	<code>filekit.h</code>
<code>fk_arc_formats</code>	how many there are (out: up to max)	<code>filekit.h</code>
<code>fk_arc_probe</code>	What the file at path is	<code>filekit.h</code>
<code>fk_arc_is_name</code>	1: its extension is a format's that is read	<code>filekit.h</code>
<code>fk_arc_open</code>	any format read; 0: err says why	<code>filekit.h</code>
<code>fk_arc_new</code>	a new one, not on the card yet ("ZIP"; 0: not a format written)	<code>filekit.h</code>
<code>fk_arc_format</code>	"ZIP" ...	<code>filekit.h</code>
<code>fk_arc_path</code>		<code>filekit.h</code>
<code>fk_arc_comment</code>	the archive's own ("": none)	<code>filekit.h</code>
<code>fk_arc_writable</code>	1: it can be changed	<code>filekit.h</code>
<code>fk_arc_method_name</code>	entry i: "Deflate", "Store, crypted"...	<code>filekit.h</code>
<code>fk_arc_test</code>	entry i read and checked, nothing written -> 0 / -1	<code>filekit.h</code>
<code>fk_extract</code>	(a type)	<code>filekit.h</code>
<code>fk_arc_extract_with</code>	0 / -1 (fk_zip_error)	<code>filekit.h</code>
<code>fk_arc_extract_bytes</code>	what that selection weighs	<code>filekit.h</code>

Name	What it does	Header
fk_arc_add	Changing an archive that can be (fk_arc_writable)	filekit.h
fk_arc_add_conflicts	names that exist already	filekit.h
fk_arc_add_bytes	the work's size	filekit.h
fk_arc_delete		filekit.h
fk_arc_rename	a file, or a folder and what is under it	filekit.h
fk_arc_new_folder		filekit.h
fk_arc_write_empty	a new archive put on the card with nothing in it	filekit.h
fk_zipmem_count	its entries, -1: not a ZIP	filekit.h
fk_zipmem_entry	Entry i	filekit.h
fk_zipmem_get	The entry of that name inflated (a 0 byte after its end) -> 0 and *out (fk_free), -1	filekit.h
fk_zipbuf_new		filekit.h
fk_zipbuf_add	0 / -1	filekit.h
fk_zipbuf_finish	the archive (fk_free) -> 0 / -1; b freed	filekit.h
fk_zipbuf_free	(given up before finish)	filekit.h
fk_exists	1 a file, 2 a folder, 0	filekit.h
fk_file_size	-1: no such file	filekit.h
fk_load	the whole file (a 0 byte after it; fk_free) -> 0 / -1	filekit.h
fk_save	0 / -1	filekit.h
fk_mkdirs	every folder of the path made -> 0 / -1	filekit.h
fk_copy	0 / -1	filekit.h
fk_move	renamed, else copied then removed	filekit.h
fk_remove	a file, or a folder and all it holds -> 0 / -1	filekit.h
fk_tree_size	The bytes under a path (a file's size)	filekit.h
fk_path_name	after the last '/' or ':' (inside path)	filekit.h
fk_path_ext	"png" (lower case, without the dot; "": none)	filekit.h

Name	What it does	Header
<code>fk_path_folder</code>	"SD:/a/b.txt" -> "SD:/a"; "SD:/a" -> "SD:/"	<code>filekit.h</code>
<code>fk_path_join</code>		<code>filekit.h</code>
<code>fk_path_unique</code>	a name not taken: "a.txt" -> "a (2).txt"	<code>filekit.h</code>
<code>fk_human_size</code>	"12.4 KB"	<code>filekit.h</code>
<code>fk_dos_time_str</code>	"28/09/2026 14:12"	<code>filekit.h</code>
<code>fs_len</code>		<code>fsutil.h</code>
<code>fs_copy</code>		<code>fsutil.h</code>
<code>fs_lower</code>		<code>fsutil.h</code>
<code>fs_ci_cmp</code>		<code>fsutil.h</code>
<code>fs_join</code>	out = dir + "/" + name (no doubled slash).	<code>fsutil.h</code>
<code>fs_basename</code>	The last path component (points into <code>path</code>).	<code>fsutil.h</code>
<code>fs_dirname</code>	Parent directory of path into out ("SD:/a/b" -> "SD:/a", "SD:/a" -> "SD:/").	<code>fsutil.h</code>
<code>fs_is_dir</code>		<code>fsutil.h</code>
<code>fs_exists</code>		<code>fsutil.h</code>
<code>fs_copy_file</code>		<code>fsutil.h</code>
<code>fs_skip_dot</code>		<code>fsutil.h</code>
<code>fs_copy_tree</code>	Copy a file or a whole folder tree (depth <= 12).	<code>fsutil.h</code>
<code>fs_remove_tree</code>	Delete a file or a whole folder tree (depth <= 12).	<code>fsutil.h</code>
<code>fs_unique_name</code>	A free path in dir for name	<code>fsutil.h</code>
<code>fs_text_fix</code>	A text file saved by a Windows editor	<code>fsutil.h</code>

filekit/filekit.h

filekit.h -- FileKit: what programs do with files, one copy for the whole system (SD:/lib/filekit.so; the shared libraries: docs/03 section 5.6). A program links lib/filekit.imp.a and calls plain C functions (fk_*):

```
compression  zlib's deflate and inflate (raw, zlib or gzip wrapped), CRC-32, Adler-32
archives     the formats known asked (fk_arc_formats), an archive of any of them opened,
extracted
tar.gz /      with choices, changed (add, delete, rename) -- ZIP read and written, tar /
              gzip read
ZIP          an archive on the card read (its entries, one read to memory or extracted, all
of
```

```

        them) and made (files and folders of the card, buffers) -- the Archiver's
engine:
        ZIP64, old code pages, ZipCrypto passwords;
        files      an archive IN MEMORY found into and built (the office formats, OpenRaster)
        folders made one loaded whole, saved; a file or a tree copied, moved, removed, measured;
        paths      the name, the extension, the folder of a path; two parts joined; a size for
people

```

Everything is integer and pointers: a program built without the FPU calls it. A buffer the library returns (void **out) is freed with fk_free, never with free / delete. The first version: 2026-10-05; the second (the same day): archives of any format, asked from the library (fk_arc_*), tar and gzip read.

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```
void fk_free (void *p);           // a buffer the library returned
```

compression (zlib)

```

#define FK_RAW      0           // the deflate stream alone (a ZIP entry's)
#define FK_ZLIB     1           // with zlib's header and Adler-32 (PNG, PDF)
#define FK_GZIP     2           // with gzip's header and CRC-32 (.gz, HTTP)
#define FK_AUTO     3           // (inflate only) zlib or gzip, told from the first bytes

```

```

unsigned fk_crc32 (unsigned crc, const void *data, unsigned n); // (crc: 0 at first, then
the last result)
unsigned fk_adler32 (unsigned adler, const void *data, unsigned n); // (adler: 1 at first)

```

The whole of data compressed (level 0..9, -1: the default 6) -> 0 and *out (fk_free) / *out_n, -1.

```
int fk_deflate (const void *data, unsigned n, int wrap, int level, void **out, unsigned
*out_n);
```

... and back. size_hint: the size expected (0: not known; the buffer grows) -> 0 / -1 (damaged). The result has a 0 byte after its end (a text can be used as it is).

```
int fk_inflate (const void *data, unsigned n, int wrap, unsigned size_hint, void **out,
unsigned *out_n);
```

a ZIP archive on the card

```

struct fk_zip_entry
{
    char      name[300];           // its path in the archive (UTF-8, '/' between the parts)
    unsigned long long size, packed; // the original size, what it takes in the archive
    unsigned   crc, dos_time;     // CRC-32; the DOS date and time (fk_dos_time_str)
    int        dir, encrypted;    // a folder; needs the password
    int        method;           // 0 stored, 8 deflate...
    int        reserved[4];
};
typedef struct fk_zip fk_zip;

```

A step of a long work: done of total bytes, the file being worked on (0: the same) -> 0 to go on, anything else to stop.

```
typedef int (*fk_progress) (void *user, unsigned long long done, unsigned long long total,
const char *name);
```

```
fk_zip *fk_zip_open (const char *path, char *err, int cap);    // 0: err says why
void fk_zip_close (fk_zip *z);
const char *fk_zip_error (fk_zip *z);    // the last failure's words
void fk_zip_password (fk_zip *z, const char *password);    // for its encrypted entries
int fk_zip_count (fk_zip *z);
int fk_zip_entry (fk_zip *z, int i, struct fk_zip_entry *out);    // 1 / 0
int fk_zip_find (fk_zip *z, const char *name);    // its index (letters' case
ignored), -1
int fk_zip_read (fk_zip *z, int i, void **out, unsigned *out_n);    // entry i in memory
(fk_free) -> 0 / -1
int fk_zip_extract (fk_zip *z, int i, const char *dest_path);    // entry i as that file
-> 0 / -1
```

Everything under prefix ("" or 0: the whole archive) into the folder dest_dir, the folders made, the names made safe for the card -> the files written, -1 (fk_zip_error; stopped: what was done stays).

```
int fk_zip_extract_all (fk_zip *z, const char *prefix, const char *dest_dir, fk_progress cb,
void *user);
```

A new archive (an existing file is replaced when it is closed): its entries said first, written by fk_zipw_close.

```
typedef struct fk_zipw fk_zipw;
fk_zipw *fk_zipw_create (const char *path);
int fk_zipw_add (fk_zipw *w, const char *disk_path, const char *name);    // a file, or a
folder and all it holds, as `name` (0: its own)
int fk_zipw_add_data (fk_zipw *w, const char *name, const void *data, unsigned n);    // a
buffer as an entry (copied)
void fk_zipw_level (fk_zipw *w, int level);    // 0 store, 1 fast, 6 (the default),
9
int fk_zipw_close (fk_zipw *w, fk_progress cb, void *user, char *err, int cap);    // written ->
0 / -1 (err); w freed
```

archives of any format the library knows (version 2, 2026-10-05)

The formats are asked, not assumed: a program that shows archives (the Archiver) lists them from here, so a format added to the library is one it has at once.

```
struct fk_format
{
    char name[16];    // "ZIP", "TAR", "TAR.GZ", "GZIP"
    char extensions[96];    // "zip jar docx ..." (lower case, a space between)
    char note[80];    // what can be done with it, in a few words
    int can_read, can_write;    // opened and extracted; made and changed (add, delete,
rename)
    int can_password;    // encrypted entries are read (with fk_arc_password)
    int reserved[4];
};
int fk_arc_formats (struct fk_format *out, int max);    // how many there are (out:
up to max)
```

What the file at path is: 1 and its format's name -- or 0 and why ("7z archives are not supported yet").

```
int fk_arc_probe (const char *path, char *format, int fcap, char *why, int wcap);
int fk_arc_is_name (const char *name);           // 1: its extension is a format's
that is read
```

An archive opened: the same handle as fk_zip (fk_zip_count, _entry, _find, _read, _extract, _extract_all, _password, _error and _close work on it whatever its format).

```
typedef struct fk_zip fk_arc;
fk_arc *fk_arc_open (const char *path, char *err, int cap);    // any format read; 0: err
says why
fk_arc *fk_arc_new (const char *path, const char *format);    // a new one, not on the card
yet ("ZIP"; 0: not a format written)
const char *fk_arc_format (fk_arc *a);                      // "ZIP" ...
const char *fk_arc_path (fk_arc *a);
const char *fk_arc_comment (fk_arc *a);                    // the archive's own ("": none)
int fk_arc_writable (fk_arc *a);                            // 1: it can be changed
int fk_arc_method_name (fk_arc *a, int i, char *out, int cap); // entry i: "Deflate",
"Store, crypted"...
int fk_arc_test (fk_arc *a, int i, fk_progress cb, void *user); // entry i read and checked,
nothing written -> 0 / -1
```

Extracting with choices.

```
#define FK_LAYOUT_FULL      0    // the archive's folders kept
#define FK_LAYOUT_FROM      1    // ... from the folder `from` down (what is above it
dropped)
#define FK_LAYOUT_FLAT      2    // the files alone, no folder
#define FK_EXISTS_ASK       0    // a file that exists: ask (the callback), replace, skip,
keep both
#define FK_EXISTS_REPLACE   1
#define FK_EXISTS_SKIP      2
#define FK_EXISTS_KEEP_BOTH 3
#define FK_ANSWER_REPLACE   1    // the callback's answers
#define FK_ANSWER_SKIP      2
#define FK_ANSWER_KEEP_BOTH 3
#define FK_ANSWER_REPLACE_ALL 4
#define FK_ANSWER_SKIP_ALL  5
#define FK_ANSWER_CANCEL    (-1)
struct fk_extract
{
    unsigned    size;          // sizeof (struct fk_extract)
    const char *dest;          // the folder to extract into (made)
    const char *selected;      // one byte an entry, not 0: taken (0: every entry)
    const char *from;          // FK_LAYOUT_FROM: the archive's folder shown
    int         layout, exists;
    int (*ask) (void *user, const char *path); // FK_EXISTS_ASK -> an FK_ANSWER_*
    fk_progress progress;
    void *user;                // (both callbacks')
    int         files, skipped; // out: what was done
};
int fk_arc_extract_with (fk_arc *a, struct fk_extract *x);    // 0 / -1 (fk_zip_error)
unsigned long long fk_arc_extract_bytes (fk_arc *a, const char *selected); // what that
selection weighs
```

Changing an archive that can be (fk_arc_writable): it is rewritten beside itself, then swapped in, and its entries are read again (the indexes change). -> 0 / -1 (fk_zip_error). paths: files and folders of the card, put in the archive's folder into ("": its top); keep_folders: a folder keeps its name and tree; replace: a name that exists is replaced, else skipped.

```

int fk_arc_add (fk_arc *a, const char *const *paths, int n, const char *into, int
keep_folders, int replace, int level,
    fk_progress cb, void *user, int *added);
int fk_arc_add_conflicts (fk_arc *a, const char *const *paths, int n, const char *into, int
keep_folders); // names that exist already
unsigned long long fk_arc_add_bytes (fk_arc *a, const char *const *paths, int n, const char
*into, int keep_folders, int replace); // the work's size
int fk_arc_delete (fk_arc *a, const char *selected, fk_progress cb, void *user);
int fk_arc_rename (fk_arc *a, const char *from, const char *to, fk_progress cb, void *user);
// a file, or a folder and what is under it
int fk_arc_new_folder (fk_arc *a, const char *name, fk_progress cb, void *user);
int fk_arc_write_empty (fk_arc *a); // a new archive put on the card with
nothing in it

```

a ZIP archive in memory (a document read whole: .docx, .xlsx, .odt, .ora ...)

```

int fk_zipmem_count (const void *zip, unsigned n); // its entries, -1: not a ZIP

```

Entry i: its name (cap bytes) and its size -> 1 / 0.

```

int fk_zipmem_entry (const void *zip, unsigned n, int i, char *name, int cap, unsigned
*size);

```

The entry of that name inflated (a 0 byte after its end) -> 0 and *out (fk_free), -1: none, damaged.

```

int fk_zipmem_get (const void *zip, unsigned n, const char *name, void **out, unsigned
*out_n);

```

... and one built: entries added (level 0: stored -- the "mimetype" of an OpenDocument first), then the archive's bytes.

```

typedef struct fk_zipbuf fk_zipbuf;
fk_zipbuf *fk_zipbuf_new (void);
int fk_zipbuf_add (fk_zipbuf *b, const char *name, const void *data, unsigned n, int level);
// 0 / -1
int fk_zipbuf_finish (fk_zipbuf *b, void **out, unsigned *out_n); // the archive (fk_free)
-> 0 / -1; b freed
void fk_zipbuf_free (fk_zipbuf *b); // (given up before finish)

```

files

```

int fk_exists (const char *path); // 1 a file, 2 a folder, 0
long long fk_file_size (const char *path); // -1: no such file
int fk_load (const char *path, void **out, unsigned *out_n); // the whole file (a 0
byte after it; fk_free) -> 0 / -1
int fk_save (const char *path, const void *data, unsigned n); // 0 / -1
int fk_mkdirs (const char *path); // every folder of the path made -> 0 /
-1

```

A file, or a folder and all it holds. dst is the new path itself (not the folder to put it in).

```

int fk_copy (const char *src, const char *dst, fk_progress cb, void *user); // 0 / -1
int fk_move (const char *src, const char *dst, fk_progress cb, void *user); // renamed, else
copied then removed
int fk_remove (const char *path); // a file, or a folder and all it holds
-> 0 / -1

```

The bytes under a path (a file's size; a folder's whole tree); *files, *folders: how many (0: not wanted).

```

long long fk_tree_size (const char *path, int *files, int *folders);

```

paths

```
const char *fk_path_name (const char *path);           // after the last '/' or ':'  
(inside path)  
void fk_path_ext (const char *path, char *out, int cap); // "png" (lower case, without  
the dot; "" : none)  
void fk_path_folder (const char *path, char *out, int cap); // "SD:/a/b.txt" -> "SD:/a";  
"SD:/a" -> "SD:/"  
void fk_path_join (char *out, int cap, const char *folder, const char *name);  
void fk_path_unique (char *path, int cap);           // a name not taken: "a.txt" -> "a  
(2).txt"  
void fk_human_size (unsigned long long bytes, char *out, int cap); // "12.4 KB"  
void fk_dos_time_str (unsigned dos_time, char *out, int cap); // "28/09/2026 14:12"  
  
}  
#endif
```

(the file helpers -- fs_join, fs_exists, fs_copy_tree...: fsutil.h, C++)

filekit/fsutil.h

fsutil.h -- file-system helpers shared by the file tools (File Viewer, trash, dock): path join, exists / is_dir, recursive copy and delete, a free "name copy" name. FileKit's (filekit.so): this header declares, fsutil.inc has the code (it was user/fsutil.h, a header of inline functions, until 2026-10-05). C++.

(a program for Onyx: FileKit's functions, by name. A PC build, or FS_INLINE: the code inline in the program -- no shared library there.)

```
#define FS_API extern "C"  
#define FS_API static inline  
  
#define FS_NAMELEN 72  
#define FS_PATHLEN 300  
int fs_len (const char *s);  
void fs_copy (char *d, const char *s, int cap);  
char fs_lower (char c);  
int fs_ci_cmp (const char *a, const char *b);
```

out = dir + "/" + name (no doubled slash).

```
void fs_join (char *out, int cap, const char *dir, const char *name);
```

The last path component (points into path).

```
const char *fs_basename (const char *path);
```

Parent directory of path into out ("SD:/a/b" -> "SD:/a", "SD:/a" -> "SD:/").

```
void fs_dirname (char *out, int cap, const char *path);  
bool fs_is_dir (const char *path);  
bool fs_exists (const char *path);  
bool fs_copy_file (const char *src, const char *dst);  
bool fs_skip_dot (const char *n);
```

Copy a file or a whole folder tree (depth <= 12).

```
bool fs_copy_tree (const char *src, const char *dst, int depth = 0);
```

Delete a file or a whole folder tree (depth <= 12).

```
bool fs_remove_tree (const char *path, int depth = 0);
```

A free path in dir for name: "name", then "name copy", "name copy 2", ... (the extension stays at the end: "notes copy.txt").

```
void fs_unique_name (char *out, int cap, const char *dir, const char *name, const char *tag =  
" copy");
```

A text file saved by a Windows editor: drop a UTF-8 BOM, turn UTF-16 (Notepad) into 8-bit. In place (b[n] must be writable); returns the new length.

```
int fs_text_fix (char *b, int n);
```